Machine learning for data analytics

Assignment 3: Probabilistic ML & Time Series I

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Each assignment should be submitted as a Jupyter notebook with complete code, visualizations, and written analysis.

Problem 3.1: Bayesian Linear Regression (25 points)

Implement Bayesian linear regression using PyMC:

- Use a polynomial regression problem with noisy data (use synthetic data or Advertising Dataset from Kaggle).
- Set appropriate priors for coefficients and noise.
- Sample from the posterior using MCMC.
- Visualize posterior distributions and credible intervals.
- Compare with standard maximum likelihood estimation.

Problem 3.2: Gaussian Process Regression (25 points)

Apply Gaussian Processes to function approximation:

- Use GPyTorch or scikit-learn.
- Test on synthetic 1D and 2D functions.
- Use Mauna Loa CO2 dataset or temperature time series from your home town
- Experiment with different kernels (RBF, Matern, Periodic).
- Visualize mean predictions and uncertainty bounds.
- Discuss computational complexity for large datasets.

Problem 3.3: ARIMA Modeling (25 points)

Forecast stock prices using ARIMA:

- Download daily prices for any stock (2+ years of data).
- Check for stationarity using ADF test.
- Use ACF/PACF plots to determine ARIMA order.
- Fit ARIMA model and validate using last 20% of data.
- Generate 30-day forecast with confidence intervals.

Problem 3.4: GARCH for Volatility (25 points)

Model volatility clustering in financial returns:

- Calculate daily returns for S&P 500 index.
- Fit GARCH(1,1) model to returns.
- Forecast volatility for next 20 days.
- Validate using realized volatility.